

HLT 501 Course Outline

HLT 501 — Health Sector Modelling

Course Outline

School	05 Sector Planning — Health
Level	500 (practitioner / advanced)
Hours	55 (indicative band 40–55)
Shape	5 modules · 25 sessions · 8 laboratories · 4 assignments · 1 final assessment
Delivery	Instructor-led — online, blended or in person
Prerequisites	EDIS 101 required. DAT 703 Household Survey Analysis strongly recommended. Regression at the level of a logistic model; Julia at the level of reading and modifying a script. R is optional
Assessment	Pass 60 %, with 60 % laboratory competency
Version	2.0 · reviewed 2026-08-27 · 12-month cycle

What this course is

HLT 501 teaches a health-sector model estate and the judgement required to say what each of its models does and does not establish. The estate is 29 Julia modules across 10 layers: a population-health layer implementing a 97-card catalogue in 14 domain families, a health-system engine running from demography through service demand and effective capacity to financing and DALYs, an optimisation layer, a geographic accessibility and climate-risk layer, and four integration links.

Household surveys are the data source for one of those layers, not the subject of the course. On the pilot survey 61 of the 97 catalogue cards ran and 36 were skipped, each naming a variable the survey does not carry. Four of its findings are negative and are taught as results.

The rule the course is built on. Every number has a model behind it and a label on it. Where the data cannot support the question, the estate refuses — and a refusal with a stated reason is a result.

Who it is for

Ministry of Health planners, statisticians and monitoring officers; health economists; Ministry of Finance sector analysts; national statistical office survey analysts; and modellers joining a health-sector planning team.

Data and access

No learner touches restricted microdata: every laboratory runs on the estate’s committed outputs and on public geographic layers, and no identifier and no cluster coordinate appears anywhere. Three labels carry through every figure — REAL for a committed survey estimate, SYNTHETIC for an output of the engine’s run on its demo country, REAL public infrastructure for the open geographic layers — and mixing a REAL and a SYNTHETIC figure in one statement without labelling both is an automatic fail.

Structure

Module	Title	Sessions	Laboratory	Hours
M1	Foundations: the estate, the data, the design	01–03	01–02	10.0
M2	The catalogue: fourteen families, and the trap in each	04–17	03–05	23.0
M3	The planning chain: population, burden, demand, capacity	18–20	06	6.5
M4	Where care is, and where it should be: access and siting	21–22	07	6.3
M5	What it costs, and what it does not establish	23–25	08	9.2
	Total	25	8	55.0

The 25 sessions and 8 laboratories, in teaching order, and the issue each turns on:

M1 Foundations: the estate, the data, the design

Sessions 01–03 · 10.0 h · Laboratory 01–02

- **01 What this system models** — the four layers and what each produces; fourteen population-health families and thirteen system blocks; how a run flows; the boundary between a survey estimate and a system-model figure
- **02 The data the estate runs on** — the inventory and the provenance tag; survey recodes and which weight applies; cluster displacement and what it means for every map; sources registered against sources actually present
- **03 Estimation under a complex survey design** — strata, clusters and weights; Taylor-linearised variance; the design effect and effective sample size; the subpopulation rule; what a published cell has to carry before it is publishable

- *Lab 01 The model estate and the catalogue* — mapping the estate by layer, reading a catalogue run as a contract, and triaging all 36 refusals into three kinds
- *Lab 02 Design-based estimation and the cross-check* — an estimate with its interval, denominator and disclosure status, and what agreement between two implementations does not establish

M2 The catalogue: fourteen families, and the trap in each

Sessions 04–17 · 23.0 h · Laboratory 03–05

Each family session runs the same seven steps: what it estimates; the trap; the models; the required diagnostics; what the committed runs produced; what the surveys refused; and where the family goes next.

- **04 A. Child Survival and Mortality** — neonatal to under-five death and competing risks; the children’s file holds survivors, so estimating mortality from it conditions on the outcome
- **05 B. Maternal Health and Continuum of Care** — antenatal adequacy, skilled attendance, postnatal care, continuum completion; four cards count four populations, so comparing them compares denominators, not care
- **06 C. Fertility, Family Planning and Reproductive Health** — children ever born, spacing, method choice, unmet need; a count without an exposure offset treats a 45-year-old and a 20-year-old as observed equally long
- **07 D. Child Nutrition and Growth** — stunting, wasting, underweight and the continuous z-scores; flag codes 9996–9999 are not measurements, and read as numbers they give a mean z-score in the thousands
- **08 E. Maternal and Adult Nutrition and Anaemia** — women’s BMI, anaemia and anaemia severity as an ordered outcome; biomarker measurement has its own eligibility and weight, so its denominator is not the women’s file
- **09 F. Child Morbidity and Treatment** — fever, ARI symptoms, diarrhoea, care seeking and treatment; treatment is asked only of children reported ill, so a blank is not missing — the filter defines the value
- **10 G. Immunization and Preventive Child Health** — full basic immunization, zero-dose children, DTP dropout, timeliness; zero-dose is a few per cent, so the regression rests on very few events and risks a singleton stratum
- **11 H. Malaria, HIV and Biomarker Models** — parasitaemia, the malaria–anaemia joint risk, HIV prevalence and testing; these cards need a configuration flag and a recorded ethics approval — a flag alone is not enough
- **12 I. WASH, Household Environment and Environmental Health** — improved water and sanitation, open defecation, clean cooking; service must map to the JMP ladders or it is comparable with no published figure
- **13 J. Women’s Empowerment, Gender and Violence-Related Health** — decision-making, empowerment, partner violence and service use; the violence module is restricted, has its own weight, and completed no cards at all on the pilot
- **14 K. Health Insurance, Access Barriers and Service Choice** — coverage, reported barriers, provider and facility-delivery choice; choice is unordered, and a binary hides the substitution a financing reform turns on
- **15 L. Inequality, Decomposition and Causal Policy Analysis** — concentration and Erreygers indices, Wagstaff, Oaxaca–Blinder, IV and difference-in-differences; the registry declares causal before estimation, and a failed gate returns a refusal
- **16 M. Multilevel, Spatial and Small-Area Models** — random effects, spatial Gaussian processes, Fay–Herriot and unit-level small area; the coordinates were deliberately displaced, so

buffer extraction is the rule

- **17 N. Prediction, Microsimulation and Bridge Models** — tree ensembles, a vulnerability index, microsimulation and the cross-sector bridges; the sparsest family, because most cards need a source a household survey lacks
- *Lab 03 Child survival and the maternal continuum* — rates and the decomposition, the hazard model and its declared biases, and a cascade transition whose deliverable is a diagnosis
- *Lab 04 Nutrition, morbidity and immunisation* — the wealth gradient, three scales of effect, and skip patterns and rare events
- *Lab 05 Inequality, small area and a map that does not leak* — five measures and the decomposition; shrinkage weights, exceedance ranking and the disclosure decision behind a published map

M3 The planning chain: population, burden, demand, capacity

Sessions 18–20 · 6.5 h · Laboratory 06

Every figure in this module is labelled SYNTHETIC: the engine's committed run is on a demo country.

- **18 Demography and epidemiology** — cohort-component projection, SIR, SEIR and the endemic incidence engine; every need estimate downstream follows the age structure, not the population total
- **19 Service demand, capacity and queueing** — need through visits-per-case, demand through care-seeking and price, Erlang-C waiting; nominal capacity is what a register says, effective capacity what staffing, readiness and hours allow
- **20 Workforce and supply chain** — WISN requirement, stock-flow projection with attrition, inventory and the reorder policy; the requirement inherits every demand assumption, and a stock-out is a failure demand never sees
- *Lab 06 Demography, epidemiology, demand and capacity* — one service traced from base-year population through incidence, need and demand to unmet demand, with every parameter named

M4 Where care is, and where it should be: access and siting

Sessions 21–22 · 6.3 h · Laboratory 07

- **21 Geographic accessibility** — travel time by road network and great circle, threshold coverage, gravity access and floating catchments; nearest-facility distance ignores capacity, so a nearer clinic serving ten times as many people is not the more accessible one
- **22 Optimisation** — facility siting as a mixed-integer programme with five objectives and a Pareto sweep; the objective is a normative choice, and min_cost, max_coverage, min_travel and min_inequality are four different problems on one dataset
- *Lab 07 Accessibility and facility optimisation* — four access measures compared, a siting run read against its stated objective, and the share of routed against great-circle travel times

M5 What it costs, and what it does not establish

Sessions 23–25 · 9.2 h · Laboratory 08

- **23 Financing, costing and DALYs** — service cost, the financing envelope and the gap, cost-effectiveness; deaths, YLL, YLD and DALYs averted. Unit costs and salaries are external, so a costing is a statement about them too
- **24 Climate-health risk and financial protection** — the heat-mortality J-curve and the malaria climate envelope; catastrophic spending on two definitions, and impoverishment. Both

functions turn over, and a national average hides the households it ruins

- **25 Macro, CGE and the handover** — exchange objects, the CDCGE adapter, the education link and the manifest's two refusals; it is a file handover — no two-way solve, and no currency value from a household survey
- *Lab 08 Financing, DALYs, climate, poverty and the handover* — costing a requirement and reading the gap against the parameters that generate it, both climate functions on their own supports, and a handover with its refusals written down

What a graduate can do

1. Locate any model in the estate and say which layer produced any given number.
2. Read a catalogue run as a contract between a specification and a dataset, and classify every refusal.
3. Produce a design-based estimate with its interval, design effect, effective sample size and disclosure decision.
4. Estimate child mortality from a birth history, and read a care cascade conditionally, distinguishing a real drop-out from a definition problem.
5. Fit and read every estimator family the catalogue uses, reporting the average marginal effect alongside every odds ratio.
6. Measure and decompose health inequality, and produce small-area estimates ranked by exceedance on a map that carries its uncertainty and leaks nothing.
7. Evaluate a risk model on cluster-held-out folds and decide from a decision curve whether targeting is worth doing at all.
8. Trace a service from base-year population to unmet demand naming every parameter, and distinguish nominal from effective capacity.
9. Choose the right accessibility measure for the question, and state an optimisation objective before its recommendation.
10. Cost a service requirement, read a financing gap and an ICER, and write a handover a minister can act on and a statistician can reproduce — including what it does not establish.

The twenty formal learning outcomes, and the map from each to the assessment that tests it, are in 12_LMS/learning_outcomes.md.

Assessment

Component	Weight	What it is
Knowledge quizzes	15 %	128 items across 17 modules
Laboratory exercises	30 %	8 laboratories, equally weighted, marked on model competence, interpretation, honesty about limits and communication
Applied assignments	25 %	The estate note and catalogue triage; the design-based estimate end to end; the domain note with a map; the planning chain traced

Component	Weight	What it is
Final assessment	30 %	Three hours, open book: foundations; estimation; inference, space and causality; and a handover written for a decision-maker

Pass 60 % overall and 60 % laboratory competency. Merit 70 %. Distinction 80 %. One resubmission of a single failed component within eight weeks. A laboratory is not complete until every validation check has been read and the provenance sidecar is in the output: an irreproducible result is no result.

Materials and set-up

A 253-page Technical Manual; 25 lecture decks and 8 laboratory decks; a Laboratory Manual, Instructor Guide and Learner Workbook; a 128-item quiz bank, four assignments, a final assessment and its marking guide; 79 datasets; 9 code exercises; 3 case studies; and the LMS package. The README lists every file.

Before the first laboratory, learners need the estate instantiated at D:/EDIP/models/HealthModelling. On this host an Application-Control policy blocks native package images, so every Julia command in this course passes `-pkgimages=no`. The full sequence is in the README and the Laboratory Manual's pre-flight section.

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